

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 06: Friday 28 August 2026

(Week 3)

Remark. Q05.7 to Q05.14 are leftover exercises from the Tuesday class in Week 3.

Remark. Q05.7 to Q05.9 are on Section 10.3-10.4.

Q05.10 to Q05.14 are on Section 12.1-12.3.

Q06.1 to Q06.6 are on Section 13.1-13.2. More questions on Section 13.1-13.3 will be set for Week 4.

Q05.7. For the following equations in Cartesian coordinates, rewrite them in terms of polar coordinates:

(i) $\frac{x^2}{9} + \frac{y^2}{4} = 1.$

(ii) $y^2 = 6x.$

Q05.8. A point $\mathbf{P}$ in $\mathbb{R}^2$ is such that both the Cartesian coordinates and polar coordinates of $\mathbf{P}$ are given by the same ordered pair of numbers. Determine the locus of $\mathbf{P}$. First give your answer in terms of polar coordinates, and then give also your answer in terms of Cartesian coordinates.

Q05.9. (a) For each part below, sketch the curve defined by the given polar equation (i.e. equation in polar coordinates):

(i) $r = 2(1 - \cos \theta)$;

(ii) $r = 6 \sin \theta$.

(b) For both parts (i) and (ii), rewrite the equation in Cartesian coordinates.

Q05.10. Consider the vector function

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

(i) Evaluate the derivative $\mathbf{v}'(t)$. Find also $\mathbf{v}'(2)$.

(ii) Evaluate the indefinite integral $\int \mathbf{v}(t) dt$.

(iii) Evaluate the definite integral $\int_0^1 \mathbf{v}(t) dt$.

Q05.11. Consider the curve in $\mathbb{R}^3$ given by

$\mathbf{r}(t) = (\ln t)\mathbf{i} + \frac{t-1}{t+3}\mathbf{j} + (t \ln t)\mathbf{k}$. Find parametric equations of the tangent line to the curve at the point where $t = 1$.

Q05.12. A particle moves in $\mathbb{R}^3$ such that its velocity at time t is given by

$$\mathbf{v}(t) = (6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}.$$

If the position of the particle at time $t = 0$ is $\mathbf{i} + 2\mathbf{j} + \pi\mathbf{k}$. Find the position of the particle at time $t = 1$.

[Remark: The position $\mathbf{r}(t)$ and velocity $\mathbf{v}(t)$ (as vector functions of time t) of the particle satisfy $\mathbf{r}'(t) = \mathbf{v}(t)$.]

Q05.13. Two particles move in $\mathbb{R}^3$ such that their positions at time t are given by:

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k},$$

$$\mathbf{r}_2(t) = (1 + 2t)\mathbf{i} + (1 + 6t)\mathbf{j} + (1 + 14t)\mathbf{k}.$$

(a) Do the particles collide? Justify your answer.

(b) Do their paths intersect? Justify your answer.

Q05.14. Find the arc length of the curve

$$x = 2t - 2 \sin t, \quad y = 2 - \sqrt{2} \cos t, \quad z = \sqrt{2} \cos t$$

for $0 \leq t \leq 4\pi$.

Q06.1. Consider the function $f(x, y, z) = \frac{x - y}{y^2 + z^2}$.

Evaluate $f(3, -1, 2)$ and $f(0, -\frac{1}{3}, 0)$.

Q06.2. Consider the function $f(x, y) = \frac{1}{\ln(9 - x^2 - y^2)}$.

Find and sketch the maximal domain of the function.

Q06.3. Consider the function $f(x, y) = \frac{x + y}{x - y}$.

Sketch the level curves $f(x, y) = k$ for $k = -2, 0, 1, 2$.

Q06.4. Describe the level sets of the following function geometrically:

$$f(x, y, z) = \sqrt{16 - x^2 - y^2 - z^2}.$$

Q06.5. Evaluate the following limits:

(i) $\lim_{(x,y) \rightarrow (0,\pi/4)} \sec x \tan(x + y).$

(ii) $\lim_{(x,y) \rightarrow (2,0)} \frac{\sqrt{2x - y} - 2}{2x - y - 4}.$

[Remark: For (ii), you may find the identity $a^2 - b^2 = (a + b)(a - b)$ useful.]

Q06.6. Does the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{y}{x^2}$ exist? Justify your answer.